

FuzeAgent Database Schema — Base (New) + Diff (Prev → New)

This document has two parts:

1. **Base Schema (New)** — full, current target schema with relations.
2. **Schema Diff (Prev → New)** — what changed from the previous version you had to the new design we agreed on.

Note: Namespaces use `test1.` as the schema prefix for clarity. Replace with your active schema as needed.

1) Base Schema (New)

1.1 Shared Infrastructure

ENUMs

- `entity_kind`: `organization | team | agent`
- `task_status`: `pending | in_progress | blocked | closed | closed_approved`
- `team_status`: `active | inactive`
- `conversation_status`: `running | paused | stopped`
- `tool_param_type`: free-text (constrained at app level: `string | number | boolean | json`)

Helper Function

- `tg_set_updated_at()` — `BEFORE UPDATE` trigger to set `updated_at = now()`
- `ensure_entity(p_id uuid, p_kind entity_kind)` — registers/validates `entities` row

```
-- Provided in SQL build: pgcrypto for gen_random_uuid()
CREATE EXTENSION IF NOT EXISTS pgcrypto;
```

1.1.1 `entities`

- **Columns:** `id uuid PK default gen_random_uuid()`, `kind entity_kind NOT NULL`, `created_at timestamptz default now()`
- **Purpose:** global identity registry for organizations, teams, agents (enables shared ownership for knowledge & conversations)

1.2 Organizations

1.2.1 organizations

- **Columns:** id uuid PK (FK→entities.id ON DELETE CASCADE), name text UNIQUE NOT NULL, description text, settings jsonb NOT NULL default '{}', created_at timestampz default now(), updated_at timestampz default now()
- **Triggers:** register entity on insert, cleanup on delete, updated_at on update

Relations

- organizations.id → entities.id (1:1 identifying via same UUID)
 - teams.organization_id → organizations.id (1→many)
 - org_tools.org_id → organizations.id (1→many)
 - goals.organization_id → organizations.id (1→many)
-

1.3 Teams

1.3.1 teams

- **Columns:** id uuid PK (FK→entities.id ON DELETE CASCADE), organization_id uuid NOT NULL (FK→organizations.id ON DELETE CASCADE), name text NOT NULL, description text, team_type text, color text, status team_status NOT NULL default 'active', settings jsonb NOT NULL default '{}', team_lead uuid NULL (FK→agents.id), created_at timestampz default now(), updated_at timestampz default now()
- **Uniqueness:** (organization_id, name)
- **Triggers:** register entity, cleanup on delete, updated_at, **team lead validator/default + auto-reassign on member changes**

Relations

- teams.id → entities.id
- teams.organization_id → organizations.id
- teams.team_lead → agents.id (must belong to same team_id)
- agents.team_id → teams.id
- goal_assigned_teams.team_id → teams.id
- tasks.team_id → teams.id
- team_tool_settings.team_id → teams.id

1.3.2 team_lead_history

- **Columns:** id bigserial PK, team_id uuid NOT NULL (FK→teams.id ON DELETE CASCADE), prev_lead_id uuid NULL (FK→agents.id ON DELETE SET NULL), new_lead_id uuid NULL (FK→agents.id ON DELETE SET NULL), reason text NOT NULL, changed_at timestampz default now()

- **Index:** (team_id, changed_at desc)
-

1.4 Agents

1.4.1 agent_templates

- **Columns:** id text PK, template_name text NOT NULL, category text, description text, model text NOT NULL, temperature numeric(3,2) default 0.70, top_p numeric(3,2), max_tokens int, frequency_penalty numeric(3,2), presence_penalty numeric(3,2), stop_sequences text[], backstory text, system_prompt text, docker_image text, concurrency_limit int, timezone text, locale text, log_level text, tags text[], metadata jsonb default '{}', created_at timestampz default now(), updated_at timestampz default now()

1.4.2 agent_template_env_vars (optional)

- **Columns:** template_id text NOT NULL (FK→agent_templates.id ON DELETE CASCADE), name text NOT NULL, value text, is_secret boolean default false, PK(template_id, name)

1.4.3 agents

- **Columns:** id uuid PK (FK→entities.id ON DELETE CASCADE), team_id uuid NOT NULL (FK→teams.id ON DELETE CASCADE), name text NOT NULL, role text, type text, status text check (status in ('active','idle','inactive')) default 'active', model text NOT NULL, temperature numeric(3,2) default 0.70 check (temperature between 0 and 2), top_p numeric(3,2), max_tokens int, frequency_penalty numeric(3,2), presence_penalty numeric(3,2), stop_sequences text[], backstory text, system_prompt text, docker_image text, concurrency_limit int, timezone text, locale text, log_level text, tags text[], metadata jsonb default '{}', template_id text NULL (FK→agent_templates.id), created_at timestampz default now(), updated_at timestampz default now(), last_activity timestampz, joined_date timestampz
- **Triggers:** register entity, cleanup on delete, updated_at

1.4.4 agent_env_vars

- **Columns:** agent_id uuid NOT NULL (FK→agents.id ON DELETE CASCADE), name text NOT NULL, value text, is_secret boolean default false, PK(agent_id, name)
-

1.5 Containers

1.5.1 containers

- **Columns:** `id` uuid PK, `agent_id` uuid NOT NULL (FK-agents.id ON DELETE CASCADE), `external_id` text, `provider` text NOT NULL, `docker_image` text, `config` jsonb default '{}', `last_run_at` timestamptz, `last_run_duration` interval, `created_at` timestamptz default now(), `updated_at` timestamptz default now()
 - **Triggers:** `updated_at`
-

1.6 Tools (org-level + overrides)

1.6.1 org_tools

- **Columns:** `id` uuid PK, `org_id` uuid NOT NULL (FK-organizations.id ON DELETE CASCADE), `key` text NOT NULL, `name` text NOT NULL, `description` text, `default_config` jsonb NOT NULL default '{}', `is_active` boolean NOT NULL default true, `created_at` timestamptz default now(), `updated_at` timestamptz default now()
- **Unique:** (`org_id`, `key`)
- **Triggers:** `updated_at`

1.6.2 org_tool_params

- **Columns:** `id` bigserial PK, `tool_id` uuid NOT NULL (FK-org_tools.id ON DELETE CASCADE), `name` text NOT NULL, `param_type` text NOT NULL, `required` boolean NOT NULL default false, `default_value` jsonb, **Unique**(`tool_id`, `name`)

1.6.3 team_tool_settings

- **Columns:** `team_id` uuid NOT NULL (FK-teams.id ON DELETE CASCADE), `tool_id` uuid NOT NULL (FK-org_tools.id ON DELETE CASCADE), `enabled` boolean NOT NULL default true, `config_override` jsonb, `updated_at` timestamptz default now(), **PK**(`team_id`, `tool_id`)

1.6.4 agent_tool_settings

- **Columns:** `agent_id` uuid NOT NULL (FK-agents.id ON DELETE CASCADE), `tool_id` uuid NOT NULL (FK-org_tools.id ON DELETE CASCADE), `enabled` boolean NOT NULL default true, `config_override` jsonb, `updated_at` timestamptz default now(), **PK**(`agent_id`, `tool_id`)
-

1.7 Goals & Milestones

1.7.1 goals

- **Columns:** `id` uuid PK, `organization_id` uuid NOT NULL (FK-organizations.id ON DELETE CASCADE), `title` text NOT NULL, `description` text, `priority` text check (priority in ('low','medium','high','critical')) default 'medium', `status` text check (status in ('planning','active','completed','on_hold')) default 'planning', `target_completion_date` timestampz, `progress_percentage` numeric(5,2) NOT NULL default 0, `created_at` timestampz default now(), `updated_at` timestampz default now()
- **Triggers:** `updated_at`

1.7.2 goal_assigned_teams

- **Columns:** `goal_id` uuid NOT NULL (FK-goals.id ON DELETE CASCADE), `team_id` uuid NOT NULL (FK-teams.id ON DELETE CASCADE), **PK**(`goal_id`, `team_id`)

1.7.3 milestones

- **Columns:** `id` uuid PK, `goal_id` uuid NOT NULL (FK-goals.id ON DELETE CASCADE), `title` text NOT NULL, `status` text NOT NULL default 'planned', `due_date` timestampz, `created_at` timestampz default now(), `updated_at` timestampz default now()
- **Triggers:** `updated_at`

1.8 Tasks

1.8.1 tasks

- **Columns:** `id` uuid PK, `team_id` uuid NOT NULL (FK-teams.id ON DELETE CASCADE), `agent_id` uuid NULL (FK-agents.id ON DELETE SET NULL), `milestone_id` uuid NULL (FK-milestones.id ON DELETE SET NULL), `title` text NOT NULL, `description` text, `status` task_status NOT NULL default 'pending', `priority` text check (priority in ('low','medium','high')) default 'medium', `progress_pct` numeric(5,2) NOT NULL default 0, `progress_notes` text, `completed_by` uuid NULL (FK-agents.id ON DELETE SET NULL), `approved_by` uuid NULL (FK-agents.id ON DELETE SET NULL), `created_at` timestampz default now(), `updated_at` timestampz default now(), `completed_at` timestampz
- **Rules (via trigger):**
 - if `status` ∈ { `closed`, `closed_approved` } ⇒ `completed_by` = NULL
 - if `status` ≠ `closed_approved` ⇒ `approved_by` = NULL
 - `completed_by` / `approved_by` (when set) must belong to same `team_id` as the task
 - optionally set `completed_at` when closing
- **Triggers:** `updated_at`, status guard

1.8.2 task_assignments

- **Columns:** task_id uuid NOT NULL (FK-tasks.id ON DELETE CASCADE), agent_id uuid NOT NULL (FK-agents.id ON DELETE CASCADE), inherited boolean NOT NULL default true, assigned_at timestamptz default now(), **PK**(task_id, agent_id)
-

1.9 Conversations (1:1 per owner)

1.9.1 conversations

- **Columns:** owner_id uuid PK (FK-entities.id ON DELETE CASCADE), title text, status conversation_status NOT NULL default 'running', created_at timestamptz default now(), updated_at timestamptz default now()
- **Guards:** owner kind must be in (organization, team, agent)
- **Auto-create:** recommended AFTER INSERT triggers on organizations, teams, agents
- **Triggers:** updated_at, owner-kind guard

1.9.2 conversation_messages

- **Columns:** id uuid PK, conversation_id uuid NOT NULL (FK-conversations.owner_id ON DELETE CASCADE), role text check (role in ('user','assistant')) NOT NULL, content text, status text check (status in ('sending','sent')) default 'sent', metadata jsonb, timestamp timestamptz default now()
 - **Indexes:** (conversation_id, timestamp desc, id desc)
-

1.10 Knowledge (Unified)

1.10.1 knowledge

- **Columns:** id uuid PK, owner_id uuid NOT NULL (FK-entities.id ON DELETE CASCADE), title text NOT NULL, filename text, type text check (type in ('document','link','text')) NOT NULL, mime_type text, size bigint, status text check (status in ('active','processing','error')) default 'active', upload_date timestamptz default now(), last_modified timestamptz default now(), content_preview text, tags text[] NOT NULL default '{}', source_url text, word_count int, extracted_text text, created_at timestamptz default now(), updated_at timestamptz default now()
 - **Guard:** owner kind must be in (organization, team, agent)
 - **Views:** org_knowledge, team_knowledge, agent_knowledge (simple kind filters)
-

2) Schema Diff (Prev → New)

This section captures only **what changed** compared to your previous version.

2.1 Agents

- **Removed:** config_jsonb
- **Added:** model, temperature, top_p, max_tokens, frequency_penalty, presence_penalty, stop_sequences, backstory, system_prompt, docker_image, concurrency_limit, timezone, locale, log_level, tags, metadata
- **Relations (unchanged):** team_id → teams(id), template_id → agent_templates(id)

New table: agent_env_vars

```
CREATE TABLE test1.agent_env_vars (  
  agent_id uuid NOT NULL REFERENCES test1.agents(id) ON DELETE CASCADE,  
  name      text NOT NULL,  
  value     text,  
  is_secret boolean NOT NULL DEFAULT false,  
  PRIMARY KEY (agent_id, name)  
);
```

2.2 Agent Templates

- **Renamed meaning:** template carries all agent settings **except** agent's own name.
- **Added:** template_name and all config columns mirroring agents.
- **Optional new table:** agent_template_env_vars (for default env vars)

2.3 Teams

- **Added:** team_lead uuid NULL (FK→agents.id)
- **New table:** team_lead_history
- **Behavior:** triggers to validate/default lead and auto-reassign when lead leaves/moves

2.4 Containers

- **Removed:** status
- **Added:** last_run_at timestampz, last_run_duration interval

2.5 Tasks

- **Status:** replaced/extended with enum task_status = ('pending', 'in_progress', 'blocked', 'closed', 'closed_approved')
- **Added:** completed_by uuid NULL FK→agents(id), approved_by uuid NULL FK→agents(id)
- **Behavior:** trigger clears/validates based on status; team consistency enforced

2.6 Conversations

- **Reworked to 1:1**
- **Removed columns:** team_id, agent_id

- **Added:** `owner_id` uuid PK (FK→`entities.id`)
- **Messages:** `conversation_id` now references `conversations.owner_id`
- **Guards:** owner kind restricted to `organization|team|agent`
- **Auto-create:** recommended from org/team/agent inserts

2.7 Knowledge

- **No structural change** (already unified on `owner_id`), only keep guard trigger & views

3) Foreign Key Map (Overview)

```

entities (id) ← organizations.id, teams.id, agents.id
organizations.id → teams.organization_id, org_tools.org_id,
goals.organization_id
teams.id → agents.team_id, tasks.team_id, goal_assigned_teams.team_id,
team_tool_settings.team_id
agents.id → tasks.agent_id, task_assignments.agent_id, agent_env_vars.agent_id,
agent_tool_settings.agent_id, containers.agent_id, teams.team_lead,
tasks.completed_by, tasks.approved_by
org_tools.id → org_tool_params.tool_id, team_tool_settings.tool_id,
agent_tool_settings.tool_id

goals.id → milestones.goal_id, goal_assigned_teams.goal_id
milestones.id → tasks.milestone_id

conversations.owner_id → conversation_messages.conversation_id
entities.id → knowledge.owner_id, conversations.owner_id

```

4) Notes & Recommendations

- **Auto-create conversations:** implement AFTER INSERT triggers on org/team/agent to guarantee “must have exactly one”.
- **Team lead enforcement:** triggers validate same-team and auto-default/reassign.
- **Tasks close/approve rules:** enforce via BEFORE trigger; keep API clean.
- **Env vars:** normalized tables are easier to audit and override.
- **Color:** optionally enforce hex via `CHECK (color ~* '^[0-9A-F]{3}{1,2}$')` or allow named tokens.
- **Indexes:** add trigrams on `name` columns for fuzzy search; compound time-desc indexes for paging.